University of Information Technology – VNU-HCM

Faculty of Computer Engineering

CSBU102 – Computer System

LAB 02

Accessing Main Memory: Words, Arrays, Bytes and Strings

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Class	ATBC26	**Practice date**	11/11/2026
Self-assessed score		**GitHub link**	https://github.com/kaquing26/CSBU102_26560070

1. LAB OVERVIEW

Purpose. In LAB 1 every value lived in one of the 32 registers. Real programs use far more data than that, such as a list of scores or a line of text, and keep it in main memory (RAM); a processor can calculate only on registers. In this session you will learn how a program finds data in memory, brings it into a register and writes a result back, for whole words (arrays) and for single bytes (characters). Every new idea is explained first and then practised.

> **Situation: a lecturer's score sheet**
>
> A lecturer keeps the scores of a class and the students' names in a small program. Even five scores and one name are too much to keep in registers, so the program stores them in main memory and moves one value at a time into a register to read, change or add it. All names and scores are fictitious.

After this session, you can:

- explain address, word and the alignment rule, and declare data with `.word`, `.byte` and `.asciiz`;
- read and write words with `lw` and `sw` (base register + offset) and compute the address of array element i as base + 4 × i, also when i is held in a register;
- read and write bytes with `lb`, `lbu` and `sb`, and explain sign extension and byte order from your own observations;
- change data in memory: swap two elements, update an element, edit characters of a string;
- predict, observe and explain what a short program does with memory, including what happens when it goes wrong.

Table 1. Structure of the session

Part	Content	Score
0	Getting started: the memory panes of MARS	–
1	Experiment 1 – Words and addresses	30
2	Experiment 2 – Arrays: reading and writing elements	30
3	Experiment 3 – Bytes, characters and strings	25
4	Wrap-up and submission	15
	Total	**100**
Ext	Extension – A 3 × 3 board in memory (optional)	+10

Working conventions

• A yellow cell is where you write an answer or a value (your text appears in red). A dashed frame is where you paste a screenshot. Listings are in grey frames; a line that starts with `# >>>` marks a place where you write code.

• Questions are labelled *Understand*, *Apply*, *Analyze* or *Evaluate*. Many activities follow PREDICT → OBSERVE → EXPLAIN: write down what you expect, run the program and record what happens, then explain any difference. A wrong prediction is never penalised; the explanation earns the marks.

• Save every program you submit as `Lab02_<StudentID>_<Part>.asm`, for example `Lab02_22520001_TN2.asm` (parts TN1 to TN3 and EXT), and start it with a comment block with your name, student ID and a one-line purpose.

PART 0. GETTING STARTED: THE MEMORY PANES OF MARS

You already know how to assemble, run and step through a program. This time you also use the Data Segment, the MARS pane that shows main memory (Table 2).

Table 2. Skills checklist

☐	Skill	Hint
☐	Check the settings	Settings: tick *Permit extended (pseudo) instructions* and *Initialize Program Counter to global 'main'*.
☐	Read the Data Segment	Each row starts with an address; the columns *Value (+0)*, *Value (+4)*, … show the words that follow. Tick *Hexadecimal Values* / *Addresses* or *ASCII* to change the display.
☐	Find the address of a label	Settings → *Show Labels Window (symbol table)*.
☐	Set a breakpoint	Tick the *Bkpt* box of an instruction in the Text Segment; Go stops just before it.

Step 0 – Quick check. Type Listing 0, assemble and run it (it prints one number read from memory). Then find the label `x` in the Labels window and the number 7 in the Data Segment.

Listing 0

```
# Listing 0 - quick check
.data
x:    .word 7
msg:  .asciiz "Value read from memory: "
nl:   .asciiz "\n"

.text
.globl main
main:
    li   $v0, 4
    la   $a0, msg
    syscall
    la   $t0, x
    lw   $a0, 0($t0)
    li   $v0, 1
    syscall
    li   $v0, 4
    la   $a0, nl
    syscall
    li   $v0, 10
    syscall
```

Text in the Run I/O pane:

```
Value read from memory: 7
```

EXPERIMENT 1. WORDS AND ADDRESSES (30 points)

Situation

Before the program can add two scores it must find them. A processor sees not a list of named scores but a very long row of numbered storage cells. This experiment shows how a program names a cell, brings its contents into a register and writes a result back.

Background: main memory, addresses and words

Main memory is a long row of **bytes** (8 bits). Every byte has its own number, its **address**. A MIPS register holds one **word** of 4 bytes, which occupies four consecutive addresses; the address of a word is the address of its first byte and must be a **multiple of 4** (the *alignment rule*). In MARS, declared data begins at `0x10010000` (Figure 1).

The processor calculates only with registers, so a program **loads** a value into a register and **stores** a result back: `lw $t0, 8($s0)` copies the word at address `$s0 + 8` into `$t0`, and `sw $t0, 8($s0)` copies `$t0` into that word. The constant 8 is the **offset** (in bytes), `$s0` is the **base register**, and base + offset is the *effective address*. A label names an address, and `la $s0, scores` copies the address of `scores` into `$s0`. Table 3 lists the directives that place data in memory.

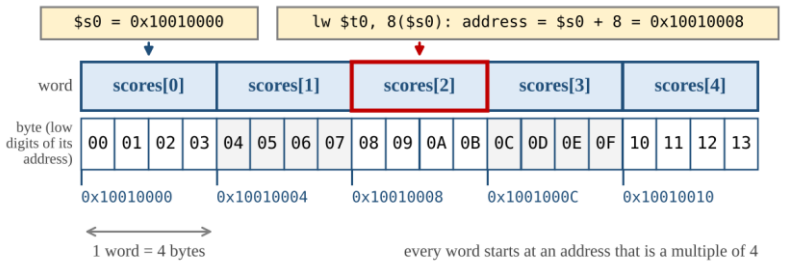


Figure 1. An array of five words in memory and the effective address of lw $t0, 8($s0)

Table 3. Data declarations

Directive	Example	What is placed in memory
`.word`	`a: .word 40, 55, 70`	Consecutive 32-bit words (an array); the label is the address of the first one.
`.byte`	`b: .byte 65`	One byte with the value 65.
`.asciiz`	`s: .asciiz "Hi"`	One byte per character, then one extra byte of value 0 that marks the end of the string.

Step 1.1 – Variables in memory (10 points)

In LAB 1, `a = b + c` was a single `add` on registers. Now the variables live in memory. Type Listing 1, assemble it, PREDICT the values in Table 4, then run it (decimal display) and OBSERVE. Also read the addresses in Table 5 (`pc` and `$sp` from the Registers pane right after assembling).

Listing 1

```
# Listing 1 - varA = varB + varC, with all three variables in memory
.data
varB: .word 25
varC: .word 17
varA: .word 0

.text
.globl main
main:
    la   $t0, varB         # $t0 = address of varB
    lw   $t1, 0($t0)       # $t1 = varB
    lw   $t2, 4($t0)       # $t2 = varC
    add  $t3, $t1, $t2     # $t3 = varB + varC
```

```
    sw   $t3, 8($t0)       # varA = $t3
    li   $v0, 10
    syscall
```

Table 4. Predict and observe Listing 1

Item	PREDICT (decimal)	OBSERVE (decimal)
`$t3`	42	42
Word at label `varA` in the Data Segment	42	42

Table 5. Addresses in the program

Item	Address (hex)
`pc` before the program runs	0x00400000
Label `varB` / `varC` / `varA`	0x10010000 / 0x10010004 / 0x10010008
`$sp` before the program runs	0x7fffeffc

Question 1.1 (Understand) Why do the addresses of `varB`, `varC` and `varA` differ by that number of bytes? The instructions, the declared data and the stack are kept in three different regions: which address belongs to which region, and which region has the highest addresses?

- Each variable is declared as .word, which is 32 bits (4 bytes) in size. Because memory is byte-addressable, consecutive words are placed 4 bytes apart.
- Regions and Address Hierarchy:
    pc (0x00400000) belongs to the Text Segment (code instructions).
    varB, varC, varA (0x10010000) belong to the Static Data Segment.
    $sp (0x7fffeffc) belongs to the Stack Segment.
+ Highest address region: The Stack Segment has the highest addresses (starting near 0x7fffffff and growing downward).

Step 1.2 – Base plus offset (10 points)

Type Listing 2 (an array of five scores). For each instruction in Table 6, work out the effective address and PREDICT the value that ends up in the destination register; run the program and record the observed value. Row (4) is an `addi`: its result is an address (hexadecimal).

Listing 2

```
# Listing 2 - addresses and offsets
.data
scores: .word 40, 55, 70, 85, 100

.text
.globl main
main:
    la   $s0, scores       # $s0 = address of scores[0]
    lw   $t0, 0($s0)       # (1)
    lw   $t1, 4($s0)       # (2)
    lw   $t2, 16($s0)      # (3)
    addi $s1, $s0, 8       # (4) $s1 = $s0 + 8
    lw   $t3, 0($s1)       # (5)
    lw   $t4, -4($s1)      # (6)
    li   $v0, 10
    syscall
```

Table 6. Effective addresses and loaded values

#	Instruction	Effective address (hex)	PREDICT value	OBSERVE value
(1)	`lw $t0, 0($s0)`	0x10010000	40	40
(2)	`lw $t1, 4($s0)`	0x10010004	55	55
(3)	`lw $t2, 16($s0)`	0x10010010	100	100
(4)	`addi $s1, $s0, 8`	–	0x10010008	0x10010008
(5)	`lw $t3, 0($s1)`	0x10010008	70	70
(6)	`lw $t4, -4($s1)`	0x10010004	55	55

Question 1.2 (Apply) Write the formula for the address of element `i` of an array of words in terms of the address of element 0, and use it for `scores[3]`. Why does 4 appear in the formula, and not 1?

Formula: $\text{Address}(\text{scores}[i]) = \text{Address}(\text{scores}[0]) + 4 \times i$
For scores[3]: $\text{Address}(\text{scores}[3]) = 0\text{x}10010000 + 4 \times 3 = 0\text{x}10010000 + 12 = 0\text{x}1001000\text{C}$
Why 4 appears instead of 1: Memory is byte-addressed, meaning every single address identifies one byte. Because each element in a word array is 4 bytes wide (32 bits), moving to index i requires advancing by $4 \times i$ bytes in memory.

Step 1.3 – Break it on purpose (10 points)

For each change in Table 7, PREDICT what will happen, change instruction (2) of Listing 2 (one change at a time), run the program and OBSERVE. In (a), also note the message that MARS shows. Restore the program afterwards.

Table 7. Controlled experiments

Change	PREDICT	OBSERVE
(a) `lw $t1, 2($s0)`	Runtime exception / crash due to unaligned word access.	Runtime exception at 0x00400008: fetch address not aligned on word boundary 0x10010002.
(b) `lw $t1, 20($s0)`	The instruction executes normally without an error, loading whatever data happens to follow the array in memory (e.g., 0).	Executes successfully; $t1 receives 0 (or the garbage/data word placed immediately after scores[4]).

Question 1.3 (Analyze) Explain (a) and (b). Which one does the processor detect, and which does it not? What does this say about who is responsible for keeping an access inside an array?

- In (a), the MIPS hardware requires that 32-bit word accesses must be aligned to addresses divisible by 4 (bottom 2 bits must be 00). Address 0x10010002 ends in...10 and causes the processor/MARS to raise an unaligned memory access exception.

- In (b) 0x10010014 is out of bounds for an array of 5 elements, but it is still a valid and aligned word address in the user's allocated data memory space.

- Detection & Responsibility The processor detects hardware alignment errors. But it doesn't track high-level software array boundaries, and will not catch out-of-bounds accesses unless the program tries to access protected/unmapped memory pages. Boundary checking is totally up to the programr / software compiler.

EXPERIMENT 2. ARRAYS: READING AND WRITING ELEMENTS (30 points)

Situation

The lecturer finds a mistake in the score sheet. Two scores must be exchanged, one score must be corrected, and the total of the five scores must be recalculated. All of this happens in memory, element by element.

Background: the address of element i

The elements of an array are words stored one after another, so element `i` is at **base + 4 × i** (Experiment 1). When the index is a constant, you write 4 × i yourself as the offset of `lw` or `sw`. When the index is held in a register, the program must compute the address: `sll $t1, $s1, 2` shifts `$s1` left by two bits, which multiplies it by 4, and `add $t1, $t1, $s0` adds the base address. Then `lw` or `sw` uses `0($t1)`.

Step 2.1 – Swap two scores (10 points)

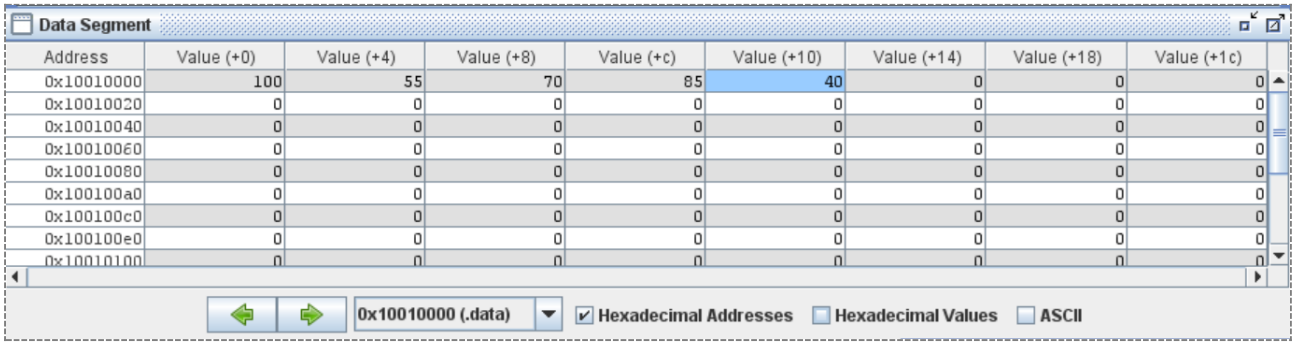
Complete Listing 3 so that it exchanges `scores[0]` and `scores[4]` in memory, using only `lw` and `sw`. After the run, the Data Segment must show 100, 55, 70, 85, 40 (decimal).

Listing 3

```
# Listing 3 - swap two scores in memory
.data
scores: .word 40, 55, 70, 85, 100

.text
.globl main
main:
    la   $s0, scores      # $s0 = address of scores[0]
    # >>> load scores[0] into $t0
    # >>> load scores[4] into $t1
    # >>> store $t1 into scores[0]
    # >>> store $t0 into scores[4]
    li   $v0, 10
    syscall
```

Screenshot of the Data Segment after the run:



Question 2.1 (Analyze) Why did you need two registers? Could one temporary register be enough? Explain.

- Why two registers were necessary: By loading both elements into two different registers ($t0 and $t1), both original values are saved before either memory location is overwritten.

- Maybe one temporary register is sufficient? Yes but only if you write to a single memory location before loading the next. However, to be able to do this without overwriting and destroying the destination, an auxiliary storage location is required. If you are limited to memory only, you cannot swap using only one register directly, since you would have to store it into the second cell, which would overwrite the value in that cell before it could be read. So you need at least two holding places (registers) to do a direct memory to memory swap .

Step 2.2 – The index in a register (10 points)

Listing 4 computes the address of `scores[i]` from the index `i` in `$s1` and prints the element. Type it and run it with i = 3 to see how it works. Then, for each variant in Table 8, PREDICT the result (for (a) to (c): `$t1` in hexadecimal and `$t2`; for (d): what happens), change the program, run it and OBSERVE. Restore the original program after each variant.

Listing 4

```
# Listing 4 - the index of the element is held in a register
# Register plan:  $s0 = address of scores[0]   $s1 = i
.data
scores: .word 40, 55, 70, 85, 100
msg:    .asciiz "scores[i] = "
nl:     .asciiz "\n"

.text
.globl main
main:
    la   $s0, scores      # $s0 = address of scores[0]
    addi $s1, $zero, 3    # i = 3
    sll  $t1, $s1, 2      # $t1 = 4 * i
    add  $t1, $t1, $s0    # $t1 = address of scores[i]
    lw   $t2, 0($t1)      # $t2 = scores[i]
    li   $v0, 4
    la   $a0, msg
    syscall
    li   $v0, 1
    add  $a0, $t2, $zero
    syscall
    li   $v0, 4
    la   $a0, nl
    syscall
    li   $v0, 10
    syscall
```

Table 8. Variants of Listing 4

Variant	PREDICT	OBSERVE
(a) `i = 0`	$t1 = 0x10010000, $t2 = 40	$t1 = 0x10010000, $t2 = 40 (prints scores[i] = 40)
(b) `i = 2`	$t1 = 0x10010008, $t2 = 70	$t1 = 0x10010008, $t2 = 70 (prints scores[i] = 70)
(c) `i = 4`	$t1 = 0x10010010, $t2 = 100	$t1 = 0x10010010, $t2 = 100 (prints scores[i] = 100)
(d) `i = 3` and `sll $t1, $s1, 1`	Shift left by 1 multiplies by 2 (3 × 2 = 6). Address becomes 0x10010006. Causes unaligned memory access exception.	Runtime exception at lw: fetch address not aligned on word boundary 0x10010006.

Question 2.2 (Analyze) Why does the shift amount have to be 2? Explain what happened in (d) and which part of the computer detected the problem.

Why is the shift amount 2: In binary, shifting left by n bits is equivalent to multiplying an integer by 2^n. Shifting by 2 multiplies by 2^2=4 which turns the element index into a byte offset .
- What happened in (d): The shift left by 1 multiplied the index 3 by 2 and this resulted in an offset of 6 bytes (0x10010000 + 6 = 0x10010006).
- Detection: The MIPS processor hardware (memory alignment checker) detected the exception. Word operations (lw) require that the two least significant address bits be 0.

Step 2.3 – Your own data (10 points)

Let D be the last digit of your student ID. In Table 9, find the index k and the new score V. Write a program, starting from Listing 4, that (1) puts k in `$s1` and V in a register and stores V into `scores[k]` with `sw`, using the address computation of Step 2.2; (2) adds up the five scores in `$s3`, loading each one with `lw` and a constant offset; (3) prints `Total:` followed by the total and a new line. First run it without step (1): the total of the original scores is 350. Then PREDICT the total for your own data by hand, run the program and complete Table 10.

Table 9. Personal data (D = last digit of the student ID)

D	k	V	D	k	V
0	0	15	5	0	62
1	1	28	6	1	77
2	2	33	7	2	84
3	3	46	8	3	91
4	4	59	9	4	98

Table 10. Results for your own data

My D	Result	Total	scores[k] in the Data Segment
0	**PREDICT**	325	15
	OBSERVE	325	15

EXPERIMENT 3. BYTES, CHARACTERS AND STRINGS (25 points)

Situation

The score sheet also contains students' names. Text is stored one character per byte, so the program must read and write single bytes, not only whole words.

Background: characters, bytes and byte order

A character is stored as a small number, its **ASCII code** (Table 18): `A` is 65 (`0x41`), `a` is 97 (`0x61`), the digit `0` is 48. A string declared with `.asciiz` is a sequence of codes followed by a byte of value 0. Three instructions work on single bytes (Table 11); a byte address can be any number.

A word occupies four consecutive bytes, so a computer must decide which byte of the number goes to the lowest address. In **little-endian** order the least significant byte comes first; in **big-endian** order the most significant byte comes first. Both are used in real computers, and the lecture slides show both. Step 3.2 shows which one MARS uses.

Table 11. Instructions that move single bytes

Instruction	Example	Meaning
`lb`	`lb $t0, 1($s0)`	Load the byte at address `$s0 + 1`; the upper 24 bits are filled with copies of the byte's top bit (sign extension).
`lbu`	`lbu $t0, 1($s0)`	Load the byte at address `$s0 + 1`; the upper 24 bits are filled with zeros.
`sb`	`sb $t0, 1($s0)`	Store the lowest 8 bits of `$t0` in the byte at address `$s0 + 1`.

Step 3.1 – The bytes of a string (6 points)

Type Listing 5 and assemble it; tick *ASCII* in the Data Segment. Using Table 18, PREDICT the hexadecimal code of each byte of `"Hi!"`; then run and complete Table 12 and the answer below it.

Listing 5

```
# Listing 5 - the bytes of a string
.data
word1: .asciiz "Hi!"

.text
.globl main
main:
    la   $s0, word1       # $s0 = address of the first byte
    lb   $t0, 0($s0)
    lb   $t1, 1($s0)
    lb   $t2, 2($s0)
    lb   $t3, 3($s0)
    li   $v0, 11          # service 11: print one character
    add  $a0, $t1, $zero  # the byte that was loaded into $t1
    syscall
    li   $v0, 10
    syscall
```

Table 12. The bytes of word1

Offset	Address (hex)	PREDICT byte (hex)	OBSERVE byte (hex)	Character
0	0x10010000	0x48	0x48	'H'
1	0x10010001	0x69	0x69	'i'
2	0x10010002	0x21	0x21	'!'
3	0x10010003	0x00	0x00	'\0'

PREDICT – character printed in the Run I/O pane:

i

Question 3.1 (Understand) How many bytes does `"Hi!"` occupy, and why one more than the number of visible characters? What is the address of the last byte?

Totals 4 bytes . " Hi ! "
• Why one more byte The directive .asciiz automatically adds one more null-terminator byte (0x00) to indicate the end of the string.
• Address of last byte: 0x10010003.

Step 3.2 – Which byte comes first? (6 points)

Listing 6 stores the word `0x41424344` and reads its four bytes one by one. Decide which byte order you expect MARS to use and PREDICT the four bytes; then run it and complete Table 13.

Listing 6

```
# Listing 6 - one word, four bytes
.data
w: .word 0x41424344

.text
.globl main
main:
    la   $s0, w
    lb   $t0, 0($s0)      # byte at offset 0
    lb   $t1, 1($s0)      # byte at offset 1
    lb   $t2, 2($s0)      # byte at offset 2
    lb   $t3, 3($s0)      # byte at offset 3
    lw   $t4, 0($s0)      # the whole word
    li   $v0, 10
    syscall
```

Table 13. Reading the bytes of one word

Item	PREDICT (hex)	OBSERVE (hex)
Byte at offset 0 (`$t0`)	0x00000044	0x00000044
Byte at offset 1 (`$t1`)	0x00000043	0x00000043
Byte at offset 2 (`$t2`)	0x00000042	0x00000042
Byte at offset 3 (`$t3`)	0x00000041	0x00000041
Whole word (`$t4`)	0x41424344	0x41424344

Question 3.2 (Analyze) Which byte of `0x41424344` is at the lowest address in MARS, and which byte order is that? Does it match your prediction? Why does `$t4` still hold `0x41424344`, and why does a program that uses only `lw` and `sw` never notice the byte order?

- 0x44 (the least significant byte) is in the lowest address This confirms MARS has Little-Endian byte order.

- $t4 still has 0x41424344 because lw reassembles the 32-bit integer according to the endianness rule of the CPU: the byte at offset 0 goes to the least-significant register byte, and the byte at offset 3 goes to the most-significant register byte.

- This is why programs that use only lw/sw never see it. Word operations write and read back 32-bit registers symmetrically. Internal byte arrangements become visible only when individual bytes are accessed via byte-level instructions (lb, sb).

Step 3.3 – Signed and unsigned bytes (6 points)

Listing 7 stores one byte of value `0xF0` and loads it twice. PREDICT `$t0` and `$t1` in hexadecimal and decimal, run it, and complete Table 14.

Listing 7

```
# Listing 7 - lb and lbu
.data
b1: .byte 0xF0

.text
.globl main
main:
    la   $s0, b1
    lb   $t0, 0($s0)
    lbu  $t1, 0($s0)
    li   $v0, 10
    syscall
```

Table 14. lb and lbu compared

Instruction	PREDICT (hex)	PREDICT (dec)	OBSERVE (hex)	OBSERVE (dec)
`lb $t0, 0($s0)`	0xFFFFFFF0	-16	0xFFFFFFF0	-16
`lbu $t1, 0($s0)`	0x000000F0	240	0x0000000F0	240

Question 3.3 (Analyze) Explain the difference in terms of sign extension (compare LAB 1, Question 2.4). Which instruction would you use to read the brightness of a pixel stored as a number from 0 to 255, and why?

- Sign extension explanation: The byte 0xF0 has a leading sign bit of 1 (1111 0000_2). lb treats the byte as a signed 8-bit integer and extends bit 7 to fill the upper 24 bits with 1s (producing 0xFFFFFFF0 = −16). lbu treats it as unsigned and zero-extends the upper 24 bits with 0s (producing 0x000000F0 = 240).
- Pixel brightness (0 to 255): Use lbu. Pixel brightness is strictly non-negative ([0,255]). Using lb would incorrectly interpret brightness values above 127 as negative numbers.

Step 3.4 – Capital letters by hand (7 points)

A capital letter has a code 32 below the matching lower-case letter (Table 18). Complete Listing 8 so that it changes the letters at offsets 0, 7 and 11 of `str1` into capital letters: for each one, load the byte with `lb`, subtract 32 and store it back with `sb`. The program must print `Nguyen Van An`. Then change `.asciiz` to `.ascii` in the declaration of `str1`, PREDICT the output, run it and OBSERVE (Table 15). Restore `.asciiz` afterwards.

Listing 8

```
# Listing 8 - capital letters at chosen positions of a string
# Register plan:  $s0 = address of the first byte of str1
.data
str1: .asciiz "nguyen van an"
tag:  .asciiz " [end]"
nl:   .asciiz "\n"

.text
.globl main
main:
    la   $s0, str1          # $s0 = address of str1
    # >>> make the byte at offset 0 a capital letter
    # >>> make the byte at offset 7 a capital letter
    # >>> make the byte at offset 11 a capital letter
    li   $v0, 4             # service 4: print the string that starts at $a0
    add  $a0, $s0, $zero
    syscall
    li   $v0, 4
    la   $a0, nl
    syscall
    li   $v0, 10
    syscall
```

Screenshot of the Run I/O pane (with `.asciiz`):

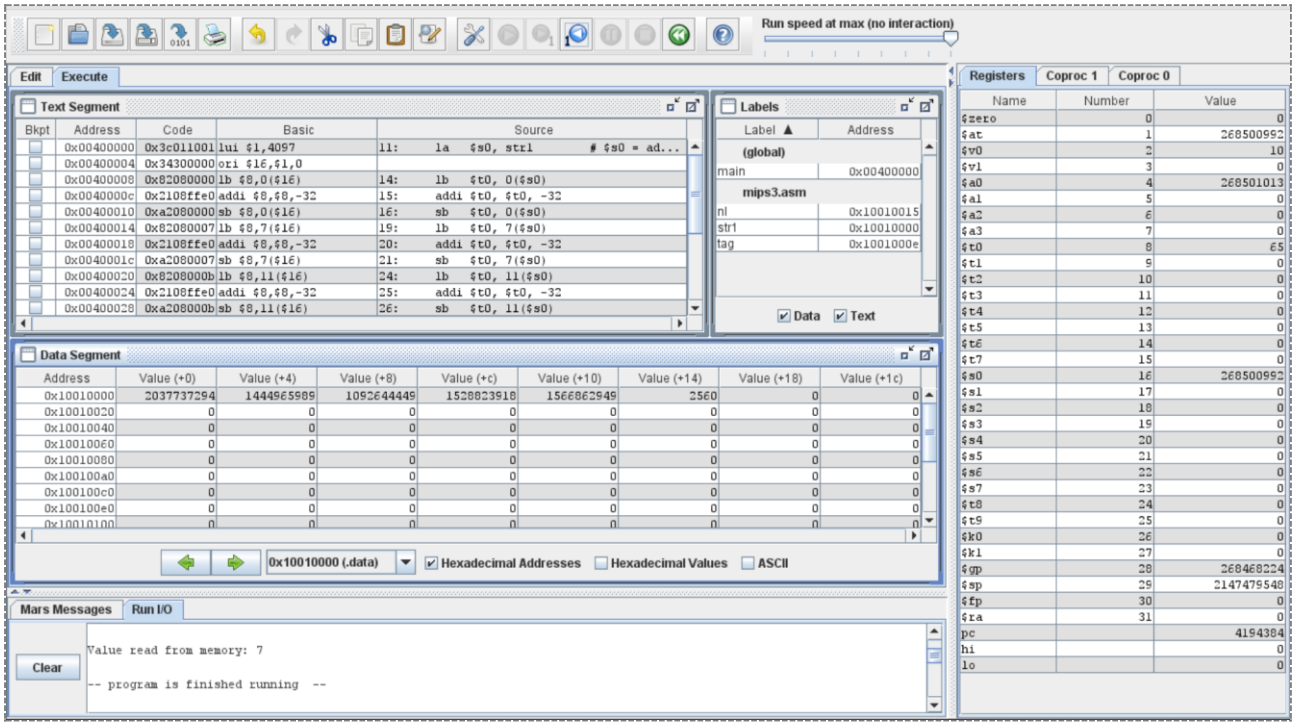


Table 15. A string without the terminating zero

Change	PREDICT	OBSERVE
`.ascii` instead of `.asciiz`	Prints Nguyen Van An [end]	Nguyen Van An [end]

Question 3.4 (Analyze) Explain the output you observed with `.ascii`. Where did the extra text come from, and how does the print service know where a string ends?

.ascii does not add a null-terminating zero (0x00) to the end of str1.

Syscall 4 prints characters starting at $a0 until it encounters a 0x00 byte. str1 does not have one . Printing goes straight into the next memory data segment label (tag) , stopping at the null terminator terminating tag .

EXTENSION. A 3 × 3 BOARD IN MEMORY (optional – up to 10 bonus points)

A game board with 3 rows and 3 columns is stored in memory as nine words, row by row: `board: .word 0, 0, 0, 0, 0, 0, 0, 0, 0`. The cell in row `r` and column `c` (both counted from 0) is word number 3 × r + c, so its address is `board` + 4 × (3 × r + c). Write a program that keeps `r` in `$s1` and `c` in `$s2`, computes this address with `sll` and `add` (3 × r is r + 2 × r), and stores the value 1 (mark X) into the cell (1, 2) and the value 2 (mark O) into the cell (2, 0). After the run, the Data Segment must show 1 and 2 in two of the nine words and 0 in all the others.

Question E.1 (Analyze) Which two words of `board` changed, at which addresses, and why does the formula give these words?

- Cell (1, 2): Index $= 3 \times 1 + 2 = 5$ (the 6th word). Address is 0x10010000 + 4 * 5 = 0x10010014. Value is 1.
- Cell (2, 0): Index $= 3 \times 2 + 0 = 6$ (the 7th word). Address is 0x10010000 + 4 * 6 = 0x10010018. Value is 2.
- Why the formula works: In row-major layout, each full row contains 3 words. To reach row r, skip r entire rows ($3 \times r$ words), then advance c individual columns to pinpoint cell (r, c). Multiplying by 4 translates word indices into exact byte offsets.

PART 4. WRAP-UP AND SUBMISSION (15 points)

Question 4.1 (Understand) Explain why `varA = varB + varC` needs two loads and one store in addition to the `add` when the variables are in memory. Use the terms *register*, *main memory*, *load* and *store*. (5 points)

The MIPS processor follows a load-store architecture: arithmetic instructions (such as add) operate exclusively on data held in **registers**, not directly on **main memory**.

1. The CPU executes two **load** instructions (lw) to transfer values from **main memory** (varB and varC) into temporary **registers**.
2. The processor computes the sum with an add instruction operating on those **registers**.
3. A **store** instruction (sw) copies the result from the target **register** back into **main memory** at varA.

Question 4.2 (Apply) Write the MIPS instructions for `scores[3] = scores[1] + scores[2]`. Assume `$s0` holds the address of `scores[0]` and use only `$t` registers. (5 points)

```
lw   $t1, 4($s0)     # $t1 = scores[1] (offset = 1 * 4)
lw   $t2, 8($s0)     # $t2 = scores[2] (offset = 2 * 4)
add  $t3, $t1, $t2   # $t3 = scores[1] + scores[2]
sw   $t3, 12($s0)    # scores[3] = $t3 (offset = 3 * 4)
```

Question 4.3 (Evaluate) `lw` and `sw` need an address that is a multiple of 4, but `lb` and `sb` accept any address. Explain why, and name one task from this LAB where a byte instruction is the right choice and one where a word instruction is. (5 points)

- Why alignment differs: A word comprises 4 bytes and must be aligned to addresses divisible by 4 so the memory interface can fetch it in a single bus cycle without requiring multi-word stitching. A byte is the smallest unit of addressable storage in MIPS, so any integer address addresses a valid individual byte.
- Appropriate task for byte instructions: Reading or altering single ASCII characters in a text string (such as capitalizing letters in Step 3.4).
- Appropriate task for word instructions: Loading and computing student test scores or integer totals (such as swapping and summing elements in Step 2.1 and Step 2.3).

Table 16. Submission checklist

☐	Item
☐	The files `Lab02_<StudentID>_TN1.asm` to `_TN3.asm` (and `_EXT.asm` if you did the extension) are saved with the correct names, begin with a comment block and assemble without errors.
☐	Every `# >>>` marker is replaced by your own code and the register plan is documented in comments.
☐	This document is complete: all yellow cells are filled and every dashed frame contains a screenshot.
☐	The `.asm` files are pushed to your GitHub repository, the link is written on the first page, and everything is submitted as announced by your instructor.

APPENDIX. REFERENCE

Table 17. Instructions and system calls used in this LAB

Instruction	Example	Meaning
`lw / sw`	`lw $t0, 8($s0)`	`$t0` = word at address `$s0 + 8` / word at that address = `$t0` (address a multiple of 4)
`lb / lbu`	`lb $t0, 1($s0)`	`$t0` = byte at address `$s0 + 1`, sign-extended / filled with zeros
`sb`	`sb $t0, 1($s0)`	byte at address `$s0 + 1` = lowest 8 bits of `$t0`
`la, li`	`la $s0, label`	pseudo-instructions: `$s0` = address of `label`; `li $t0, 7`: `$t0` = 7
`add, addi`	`addi $t0, $t1, 20`	`$t0` = `$t1` + 20 (`add` uses two registers)
`sll`	`sll $t1, $s1, 2`	`$t1` = `$s1` shifted left by 2 bits (4 × `$s1`)
`syscall`	`$v0 = 1, 4, 10`	print integer (`$a0`); print string (`$a0` = address); exit

Table 18. ASCII codes used in this LAB

Character	Decimal	Hexadecimal	Remark
end of string	0	`0x00`	terminator of `.asciiz`
space; `!`	32; 33	`0x20; 0x21`	
digits `0` to `9`	48 – 57	`0x30 - 0x39`	consecutive
capital letters `A` to `Z`	65 – 90	`0x41 - 0x5A`	consecutive
lower-case letters `a` to `z`	97 – 122	`0x61 - 0x7A`	consecutive